

A Stochastic Resource-Extraction Economy of Earth with Heterogeneous Geology

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Abstract

This paper develops the foundations of a stochastic resource-extraction economy for a heterogeneous Earth. Unlike constant-density planetary models, Earth is represented as a sphere possessing a radially varying density field. Geological uncertainty is incorporated through stochastic radius and mass distributions, while extraction prices and costs are modelled as random variables.

The framework unifies geometry, probability theory, geophysics, resource economics, and statistical modelling. We derive the extractable mass theorem for a heterogeneous Earth, characterize density distributions, and establish the basic profit-generating mechanism upon which later economic and geopolitical analysis may be constructed.

The paper ends with “The End”

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Part I

1 Introduction

Resource extraction has shaped civilization from prehistory to the modern era.

Agricultural societies depended upon extraction of biological resources.

Industrial societies depended upon extraction of coal, iron, copper, and oil.

Contemporary economies increasingly depend upon extraction of semiconductors, rare earth elements, lithium, uranium, and other strategic materials.

Despite enormous literature concerning resource economics, most models begin with resource stocks as primitive quantities.

The present framework instead begins from first principles:

1. planetary geometry,
2. geological density distributions,
3. extraction accessibility,
4. stochastic uncertainty.

The resulting model provides a geometric foundation for resource economics.

2 Planetary Geometry

2.1 The Earth as a Sphere

Let Earth be represented as a sphere of radius

$$R > 0.$$

The volume enclosed within radius r is

$$V(r) = \frac{4}{3}\pi r^3.$$

Consequently,

$$V(R) = \frac{4}{3}\pi R^3.$$

2.2 Spherical Shells

For an infinitesimal radial thickness dr at radius r , the shell volume is

$$dV = 4\pi r^2 dr.$$

This expression forms the fundamental geometric element of the model.

3 A Heterogeneous Earth

3.1 Radially Varying Density

Unlike constant-density approximations, Earth is assumed to possess a density field

$$\rho(r),$$

where

$$0 \leq r \leq R.$$

Definition 1 (Density Field). *A density field is a measurable function*

$$\rho : [0, R] \rightarrow \mathbb{R}_+.$$

The density field captures geological heterogeneity.
Examples include:

- crust,
- mantle,
- outer core,
- inner core.

4 Total Planetary Mass

The total mass of Earth is obtained by integrating density over volume.

Theorem 1 (Planetary Mass Formula). *The total mass satisfies*

$$M = \int_0^R 4\pi\rho(r)r^2dr.$$

Proof. An infinitesimal shell possesses mass

$$dM = \rho(r)dV.$$

Substituting

$$dV = 4\pi r^2dr$$

gives

$$dM = 4\pi\rho(r)r^2dr.$$

Integrating over all radii yields

$$M = \int_0^R 4\pi\rho(r)r^2dr.$$

□

5 Uniform Density as a Special Case

Suppose

$$\rho(r) = \rho_0.$$

Then

$$M = 4\pi\rho_0 \int_0^R r^2dr.$$

Therefore

$$M = \frac{4}{3}\pi\rho_0R^3.$$

Solving for density,

$$\rho_0 = \frac{3M}{4\pi R^3}.$$

Thus the familiar homogeneous-Earth formula appears as a special case of the heterogeneous model.

6 Excavation Geometry

6.1 Surface-Inward Extraction

Mining and drilling naturally proceed from the surface toward the centre.
Let

$$r_{\max} \in [0, R]$$

denote the radius of the deepest remaining unexcavated core.
Then extraction removes material in the shell

$$r_{\max} \leq r \leq R.$$

6.2 Extractable Mass

Theorem 2 (General Extractable Mass Theorem). *The extractable mass is*

$$M_{\text{ext}} = \int_{r_{\max}}^R 4\pi\rho(r)r^2 dr.$$

Proof. The shell mass between radii $r_{\max}$ and R equals

$$\int_{r_{\max}}^R dM.$$

Substituting

$$dM = 4\pi\rho(r)r^2 dr$$

completes the proof. □

7 Unexcavated Mass

The remaining planetary core contains

$$M_{\text{core}} = \int_0^{r_{\max}} 4\pi\rho(r)r^2 dr.$$

Consequently,

$$M = M_{\text{core}} + M_{\text{ext}}.$$

Corollary 1. *The mass decomposition*

$$M = \int_0^{r_{\max}} 4\pi\rho(r)r^2 dr + \int_{r_{\max}}^R 4\pi\rho(r)r^2 dr$$

always holds.

8 Accessibility Fraction

Definition 2. *Define the accessibility coefficient*

$$\kappa = \frac{M_{\text{ext}}}{M}.$$

Hence

$$0 \leq \kappa \leq 1.$$

Substituting the mass expressions,

$$\kappa = \frac{\int_{r_{\max}}^R 4\pi\rho(r)r^2 dr}{\int_0^R 4\pi\rho(r)r^2 dr}.$$

Remark 1. *For heterogeneous Earths, accessibility depends not only on geometry but also on the density distribution.*

9 Uniform Density Benchmark

If

$$\rho(r) = \rho_0,$$

then

$$\kappa = \frac{\int_{r_{\max}}^R r^2 dr}{\int_0^R r^2 dr}.$$

Evaluating,

$$\kappa = \frac{R^3 - r_{\max}^3}{R^3}.$$

Therefore

$$\boxed{\kappa = 1 - \left(\frac{r_{\max}}{R}\right)^3}.$$

10 Geological Resource Value Density

Not all mass possesses identical economic value.

Define

$$v(r) \geq 0$$

as value per unit mass at depth r .

Examples include:

- iron concentration,
- copper concentration,
- uranium concentration,
- rare-earth concentration,
- hydrocarbon concentration.

Definition 3. *The recoverable resource value is*

$$V_{\text{ext}} = \int_{r_{\max}}^R 4\pi v(r)\rho(r)r^2 dr.$$

11 Probabilistic Earth

The exact values of planetary parameters are uncertain.

We therefore introduce stochastic variables.

11.1 Random Planetary Radius

Assume

$$R \sim \text{LogNormal}(\mu_R, \sigma_R^2).$$

Hence

$$\ln R \sim N(\mu_R, \sigma_R^2).$$

Expected radius:

$$E[R] = e^{\mu_R + \frac{1}{2}\sigma_R^2}.$$

11.2 Random Planetary Mass

Assume

$$M \sim \text{LogNormal}(\mu_M, \sigma_M^2).$$

Thus

$$\ln M \sim N(\mu_M, \sigma_M^2).$$

Expected mass:

$$E[M] = e^{\mu_M + \frac{1}{2}\sigma_M^2}.$$

12 Stochastic Density Fields

The density field itself may be random.

Write

$$\rho(r, \omega)$$

where

$$\omega \in \Omega$$

belongs to a probability space.

Then total mass becomes

$$M(\omega) = \int_0^R 4\pi\rho(r, \omega)r^2 dr.$$

Similarly,

$$M_{\text{ext}}(\omega) = \int_{r_{\text{max}}}^R 4\pi\rho(r, \omega)r^2 dr.$$

13 Layered Earth Approximation

A useful approximation partitions Earth into layers.

Suppose

$$0 = r_0 < r_1 < \cdots < r_n = R.$$

Within each layer,

$$\rho(r) = \rho_i.$$

Then

$$M = \sum_{i=1}^n \frac{4\pi}{3} \rho_i (r_i^3 - r_{i-1}^3).$$

This discretization connects geological modelling with computational methods.

14 Statistical Summary

Quantity	Interpretation
R	Planetary radius
M	Planetary mass
$\rho(r)$	Density field
$r_{\max}$	Radius of remaining core
M_{ext}	Extractable mass
M_{core}	Unexcavated mass
κ	Accessibility coefficient
$v(r)$	Resource value density
V_{ext}	Recoverable value

Table 1: Fundamental variables.

15 Geometric Illustration

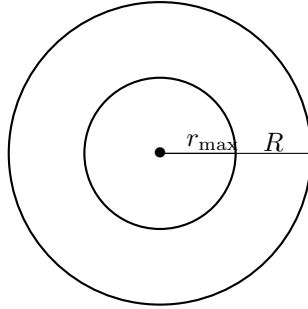


Figure 1: Geometric Illustration.

The inner circle represents the remaining planetary core while the outer shell represents the extractable region.

16 Conclusion

Beginning from planetary geometry, we constructed a stochastic resource-extraction framework for a heterogeneous Earth.

The central result is the General Extractable Mass Theorem,

$$M_{\text{ext}} = \int_{r_{\max}}^R 4\pi \rho(r) r^2 dr,$$

which reduces to

$$M \left(1 - \frac{r_{\max}^3}{R^3} \right)$$

under uniform density.

The framework developed here provides the mathematical foundation for a stochastic planetary resource economy. Subsequent parts will introduce prices, costs, profit maximization, welfare economics, finance, geopolitical competition, warfare economics, statistical estimation, and artificial intelligence.

Part II

17 Economics of a Heterogeneous Earth

17.1 Introduction

Part I established the geometric and geological foundations of a stochastic Earth possessing a radially varying density field

$$\rho(r).$$

We now construct the economic layer of the model.

The central idea is that geological resources acquire economic significance only when they can be extracted profitably.

Consequently, economics emerges from the interaction of:

1. geological abundance,
2. extraction accessibility,
3. technological capability,
4. market valuation.

18 Price and Cost Fields

18.1 Depth-Dependent Prices

Different materials possess different economic values.

Define

$$p(r) \geq 0$$

as the expected market price per unit mass recovered at depth r .

Examples include:

- iron ore,
- copper ore,
- uranium ore,
- hydrocarbons,
- rare-earth minerals.

The function $p(r)$ may vary substantially with geological depth.

18.2 Depth-Dependent Costs

Define

$$c(r) \geq 0$$

as extraction cost per unit mass at depth r .

Generally,

$$\frac{dc(r)}{dr} < 0$$

is unlikely.

More realistically,

$$\frac{dc(r)}{dr} > 0,$$

since deeper extraction is typically more difficult.

Examples include:

- drilling costs,
- ventilation costs,
- transportation costs,
- energy requirements,
- safety costs.

19 Margin Density

Define net margin density

$$m(r) = p(r) - c(r).$$

Positive values indicate profitable extraction.

Negative values indicate economically infeasible extraction.

Definition 4. *The margin field is*

$$m : [0, R] \rightarrow \mathbb{R}.$$

20 Recoverable Economic Value

The recoverable economic value of the extractable shell equals

$$V_{\text{ext}} = \int_{r_{\text{max}}}^R 4\pi p(r) \rho(r) r^2 dr.$$

Similarly, extraction cost equals

$$C_{\text{ext}} = \int_{r_{\text{max}}}^R 4\pi c(r) \rho(r) r^2 dr.$$

Theorem 3 (Value Decomposition Theorem). *The total profit satisfies*

$$\Pi = V_{\text{ext}} - C_{\text{ext}}.$$

Proof. By definition,

$$\Pi = \int_{r_{\text{max}}}^R 4\pi [p(r) - c(r)] \rho(r) r^2 dr.$$

Separating the integral gives

$$\Pi = V_{\text{ext}} - C_{\text{ext}}.$$

□

21 Fundamental Profit Equation

Substituting the margin field,

$$m(r) = p(r) - c(r),$$

yields

$$\boxed{\Pi = \int_{r_{\text{max}}}^R 4\pi m(r) \rho(r) r^2 dr.}$$

This equation forms the foundation of the planetary extraction economy.

22 Average Resource Quality

Define

$$\bar{p} = \frac{\int_{r_{\max}}^R p(r)\rho(r)r^2 dr}{\int_{r_{\max}}^R \rho(r)r^2 dr}.$$

Similarly,

$$\bar{c} = \frac{\int_{r_{\max}}^R c(r)\rho(r)r^2 dr}{\int_{r_{\max}}^R \rho(r)r^2 dr}.$$

The average margin becomes

$$\bar{m} = \bar{p} - \bar{c}.$$

Consequently,

$$\Pi = \bar{m}M_{\text{ext}}$$

whenever average-margin approximations are employed.

23 A Resource Production Function

Resource extraction may be viewed as a production process.

Let

- K denote capital,
- L denote labour,
- T denote technology.

Define extraction output

$$Q = F(K, L, T, M_{\text{ext}}).$$

A Cobb–Douglas specification is

$$Q = AK^\alpha L^\beta M_{\text{ext}}^\gamma.$$

Proposition 1. *Output elasticity with respect to extractable mass equals*

$$\gamma.$$

Proof. Differentiate logarithmically:

$$\frac{\partial \ln Q}{\partial \ln M_{\text{ext}}} = \gamma.$$

□

24 Exhaustible Resource Dynamics

Suppose extraction occurs through time.

Let

$$X(t)$$

denote cumulative extraction.

Then remaining extractable stock is

$$S(t) = M_{\text{ext}} - X(t).$$

The stock equation becomes

$$\frac{dS}{dt} = -q(t),$$

where

$$q(t) = \frac{dX}{dt}$$

is the extraction rate.

25 Hotelling-Type Scarcity Dynamics

For exhaustible resources, scarcity generates rents.

Define shadow value

$$\lambda(t).$$

The classical scarcity condition becomes

$$\frac{\dot{\lambda}}{\lambda} = r,$$

where r is the interest rate.

Thus scarcity rents grow exponentially:

$$\lambda(t) = \lambda_0 e^{rt}.$$

The heterogeneous Earth framework generalizes Hotelling's analysis because extraction costs and values vary with depth.

26 Resource Rent

Define resource rent

$$R_{\text{rent}} = \Pi - wL - r_K K,$$

where

- w is wage,
- r_K is cost of capital.

Positive rents generate incentives for exploration and geopolitical competition.

27 National Income

Suppose resource extraction contributes to GDP.

Let

$$Y = Y_O + \Pi,$$

where

$$Y_O$$

denotes output from all non-resource sectors.

Then

$$\frac{\partial Y}{\partial \Pi} = 1.$$

Resource profits therefore directly augment national income.

28 Welfare Economics

28.1 Social Welfare

Let social welfare be

$$W = U(C) - D(E),$$

where

- C denotes consumption,
- E denotes environmental damage.

Extraction increases consumption but may also increase environmental costs.

28.2 Social Planner Problem

The planner solves

$$\max_{r_{\max}} W.$$

Substituting,

$$\max_{r_{\max}} [U(C) - D(E)].$$

29 Environmental Externalities

Define environmental damage density

$$e(r).$$

Total environmental damage becomes

$$E = \int_{r_{\max}}^R 4\pi e(r) r^2 dr.$$

Net social value therefore equals

$$\Pi - E.$$

30 Intergenerational Welfare

Future generations inherit remaining resources.

The remaining stock equals

$$M_{\text{core}} = \int_0^{r_{\max}} 4\pi \rho(r) r^2 dr.$$

Consequently, extraction today imposes opportunity costs upon future generations.

The planner may maximize

$$\sum_{t=0}^{\infty} \beta^t W_t,$$

where

$$0 < \beta < 1$$

is the social discount factor.

31 Political Economy

Resource extraction creates rents.

Rents generate incentives for:

- taxation,
- lobbying,
- corruption,
- rent-seeking,
- political capture.

Define rent-seeking expenditures

$$Z.$$

Then effective resource income becomes

$$\Pi - Z.$$

32 State Capacity

Let

$$\tau$$

denote the resource tax rate.

Government revenue becomes

$$G = \tau\Pi.$$

Hence

$$\frac{\partial G}{\partial \Pi} = \tau.$$

Resource-rich states can therefore finance larger bureaucracies, militaries, and infrastructure systems.

33 Geopolitical Competition

Suppose two states compete for control of the same resource region.

Let

$$\Pi_A$$

and

$$\Pi_B$$

denote expected profits.

The strategic value of the territory equals

$$V_S = \Pi_A + \Pi_B.$$

Conflict likelihood generally increases with

$$V_S.$$

34 International Trade

Suppose extracted resources are exported.

Let

$$X_R$$

denote resource exports.

Then trade balance satisfies

$$TB = X_R + X_O - M_O,$$

where

- X_O denotes non-resource exports,
- M_O denotes imports.

Large resource exports may therefore improve external balances.

35 Resource Dependence

Define

$$\theta = \frac{\Pi}{Y}.$$

Interpretation:

- $\theta \approx 0$ implies diversification,
- $\theta \approx 1$ implies extreme dependence.

High values of θ increase vulnerability to commodity-price shocks.

36 Finance

Financial markets capitalize expected future extraction profits.

Define firm value

$$F = \sum_{t=0}^{\infty} \frac{E[\Pi_t]}{(1+i)^t}.$$

This is the discounted present value of future resource profits.

37 Resource Options

Extraction rights resemble real options.

Let

$$\Pi_t$$

be stochastic.

Management may choose whether or not to extract.

Consequently, resource ownership contains option value beyond immediate profits.

38 A Stochastic Planetary Economy

Combining Part I and Part II yields

$$\Pi = \int_{r_{\max}}^R 4\pi[p(r) - c(r)]\rho(r)r^2 dr.$$

The expression unifies:

1. geology through $\rho(r)$,
2. economics through $p(r)$,
3. technology through $c(r)$,
4. geometry through r^2 ,
5. accessibility through $r_{\max}$.

39 Summary Table

Quantity	Interpretation
$p(r)$	Price density
$c(r)$	Cost density
$m(r)$	Margin density
V_{ext}	Revenue
C_{ext}	Cost
Π	Profit
$S(t)$	Remaining stock
$q(t)$	Extraction rate
$\lambda(t)$	Scarcity rent
W	Social welfare
G	Government revenue
θ	Resource dependence ratio
F	Firm valuation

Table 2: Economic variables introduced in Part II.

40 Conclusion

Part II constructed the economic architecture of the heterogeneous-Earth framework.

The key result is the Fundamental Profit Equation

$$\Pi = \int_{r_{\max}}^R 4\pi[p(r) - c(r)]\rho(r)r^2 dr.$$

This equation links geology, technology, accessibility, and market valuation within a single mathematical framework. It provides the foundation for subsequent analysis of finance, warfare economics, geopolitics, international relations, statistics, data science, and artificial intelligence.

Part III

41 Physics of Planetary Resource Extraction

41.1 Introduction

Resource extraction is ultimately constrained by physical laws.

Every extraction process requires:

1. energy,
2. matter transport,
3. mechanical work,
4. thermodynamic transformations.

Consequently, the economics developed in Part II must be embedded within a physical framework.

42 Gravitational Structure

Let

$$M(r) = \int_0^r 4\pi\rho(s)s^2 ds$$

denote enclosed mass.

By Newtonian gravitation, gravitational acceleration at radius r satisfies

$$g(r) = \frac{GM(r)}{r^2},$$

where

$$G$$

is the gravitational constant.

42.1 Gravitational Potential

The gravitational potential satisfies

$$\Phi(r) = - \int_r^\infty g(s) ds.$$

Extraction therefore requires overcoming gravitational constraints.

43 Energy Requirements

Suppose

$$\varepsilon(r)$$

denotes energy required to extract one unit of mass from depth r .

Total extraction energy becomes

$$E_{\text{req}} = \int_{r_{\text{max}}}^R 4\pi\varepsilon(r)\rho(r)r^2 dr.$$

Theorem 4 (Energy Requirement Theorem). *Total extraction energy is proportional to extracted mass whenever*

$$\varepsilon(r) = \varepsilon_0.$$

Proof. Substituting

$$\varepsilon(r) = \varepsilon_0$$

gives

$$E_{\text{req}} = \varepsilon_0 M_{\text{ext}}.$$

□

44 Thermodynamic Constraints

The Second Law of Thermodynamics implies that extraction can never be perfectly efficient.
Let

$$0 < \eta < 1$$

denote efficiency.

Then useful energy satisfies

$$E_{\text{useful}} = \eta E_{\text{input}}.$$

Hence

$$E_{\text{input}} \geq E_{\text{req}}.$$

45 Mass-Energy Equivalence

Einstein's relation gives

$$E = mc^2.$$

Consequently, extracted mass possesses theoretical energetic value

$$E_{\text{mass}} = M_{\text{ext}} c^2.$$

This represents an upper bound rather than an economically achievable quantity.

46 Geothermal Constraints

Earth possesses an internal temperature field

$$T(r).$$

Deep extraction may become prohibitively expensive due to:

- heat,
- pressure,
- structural instability.

Consequently,

$$c(r)$$

typically rises sharply as depth increases.

47 Chemistry of Resource Extraction

47.1 Resource Concentration

Define concentration field

$$\chi(r).$$

For example:

- copper concentration,
- uranium concentration,
- lithium concentration,
- rare-earth concentration.

Recoverable quantity becomes

$$Q_\chi = \int_{r_{\max}}^R 4\pi\chi(r)\rho(r)r^2dr.$$

47.2 Chemical Recovery

Not all material is recoverable.

Let

$$0 \leq \gamma(r) \leq 1$$

denote recovery efficiency.

Recoverable resource mass becomes

$$Q_{\text{rec}} = \int_{r_{\max}}^R 4\pi\gamma(r)\chi(r)\rho(r)r^2dr.$$

48 Biology and Resource Systems

48.1 Biological Dependence on Extraction

Biological systems depend upon extracted resources.

Examples include:

- phosphorus,
- nitrogen,
- potassium,
- freshwater,
- energy resources.

Economic output therefore depends indirectly upon biological carrying capacity.

48.2 Ecological Damage

Let

$$B$$

denote biodiversity.

Suppose

$$B = B(E),$$

where

$$E$$

denotes environmental disturbance.

Typically,

$$\frac{dB}{dE} < 0.$$

Thus aggressive extraction may reduce biodiversity.

49 Psychology of Resource Economies

49.1 Expectations

Market participants form beliefs regarding future prices.

Let

$$P_t$$

denote commodity price.

Expected price becomes

$$E_t[P_{t+1}].$$

Investment decisions depend upon expectations rather than solely current prices.

49.2 Behavioral Biases

Resource markets exhibit:

- optimism bias,
- overconfidence,
- anchoring,
- herd behaviour,
- recency effects.

Consequently,

$$E_t[P_{t+1}]$$

may systematically differ from realized prices.

50 Psychiatry and Resource Shocks

Resource booms and busts influence:

- unemployment,
- migration,
- family stability,
- social cohesion.

Define stress index

$$S_t.$$

Suppose

$$S_t = f(|\Delta P_t|).$$

Large commodity shocks may therefore increase psychological stress.

51 Statistical Framework

51.1 Random Prices

Assume

$$p(r, \omega)$$

is stochastic.

Expected price field:

$$\bar{p}(r) = E[p(r, \omega)].$$

Similarly,

$$\bar{c}(r) = E[c(r, \omega)].$$

Expected profit becomes

$$E[\Pi] = E \left[\int_{r_{\max}}^R 4\pi(p - c)\rho r^2 dr \right].$$

Under integrability conditions,

$$E[\Pi] = \int_{r_{\max}}^R 4\pi E[(p - c)\rho] r^2 dr.$$

52 Variance of Profit

Profit uncertainty satisfies

$$\text{Var}(\Pi) = E[\Pi^2] - (E[\Pi])^2.$$

Risk-adjusted decisions depend upon both expected value and variance.

53 Monte Carlo Estimation

Suppose

$$\omega_1, \dots, \omega_N$$

are sampled scenarios.

Then

$$\hat{\Pi} = \frac{1}{N} \sum_{i=1}^N \Pi(\omega_i)$$

estimates expected profit.

Theorem 5 (Monte Carlo Consistency). *As*

$$N \rightarrow \infty,$$

$$\hat{\Pi} \rightarrow E[\Pi].$$

Proof. Direct application of the Law of Large Numbers.

□

54 Data Science

54.1 Parameter Estimation

Observed geological data may estimate:

- density fields,
- concentration fields,
- cost fields,
- price distributions.

Define parameter vector

$$\theta.$$

Estimation solves

$$\hat{\theta} = \arg \max_{\theta} L(\theta),$$

where

$$L(\theta)$$

is the likelihood function.

55 Machine Learning

55.1 Supervised Learning

Let

$$X$$

denote geological features.

Let

$$Y$$

denote resource value.
The learning problem becomes

$$Y = f(X) + \varepsilon.$$

Examples include:

- ore-grade prediction,
- reserve estimation,
- cost forecasting.

56 Neural Networks

A neural network approximates

$$\hat{Y} = f_{\theta}(X).$$

Training solves

$$\min_{\theta} \sum_i (Y_i - \hat{Y}_i)^2.$$

57 Reinforcement Learning

Let

- state s_t describe geological conditions,
- action a_t describe extraction decisions,
- reward R_t denote profits.

The planner solves

$$\max_{\pi} E \left[\sum_{t=0}^{\infty} \beta^t R_t \right].$$

58 World Models

A planetary extraction system may be represented as

$$s_{t+1} = F(s_t, a_t, \varepsilon_t).$$

Here

- geology evolves,
- prices evolve,
- technology evolves,
- environmental conditions evolve.

Such models naturally connect resource economics to modern AI.

59 Computer Science

59.1 Layered Approximation Algorithm

Suppose Earth is partitioned into n layers.

The extraction algorithm becomes

```
Input:
rho_i
p_i
c_i
r_i

Profit = 0

For i = 1,...,n

Mass_i = (4*pi/3) rho_i (r_i^3-r_(i-1)^3)

Profit += (p_i-c_i) Mass_i

End

Output Profit
```

60 Information Theory

Suppose uncertainty regarding geology is represented by random variable

$$G.$$

Entropy is

$$H(G) = - \sum_i p_i \log p_i.$$

Exploration activities reduce entropy and increase information.

Information therefore possesses economic value.

61 Systems Theory

The planetary extraction economy consists of interacting subsystems:

1. geology,
2. engineering,
3. markets,
4. governments,
5. ecosystems,
6. technologies.

The resulting system is adaptive, nonlinear, and dynamic.

62 The General Stochastic Extraction Equation

Combining Parts I–III yields

$$\Pi = \int_{r_{\max}}^R 4\pi[p(r) - c(r)]\rho(r)r^2 dr.$$

Expected profit becomes

$$E[\Pi] = \int_{r_{\max}}^R 4\pi E[(p - c)\rho]r^2 dr.$$

These equations unify:

- geometry,
- geology,
- economics,
- thermodynamics,
- chemistry,
- biology,
- psychology,
- statistics,
- computer science,
- machine learning.

63 Summary Table

Quantity	Interpretation
$g(r)$	Gravitational acceleration
$\Phi(r)$	Gravitational potential
$\varepsilon(r)$	Extraction energy density
η	Energy efficiency
$T(r)$	Temperature field
$\chi(r)$	Resource concentration
$\gamma(r)$	Recovery efficiency
B	Biodiversity measure
S_t	Stress index
θ	Statistical parameter vector
$H(G)$	Geological entropy

Table 3: Variables introduced in Part III.

64 Conclusion

Part III embedded the heterogeneous-Earth resource economy within the broader frameworks of physics, chemistry, biology, psychology, psychiatry, statistics, data science, artificial intelligence, information theory, and systems science.

The central conclusion is that resource extraction is not merely an economic activity. It is a coupled physical, chemical, biological, informational, and computational process operating on a stochastic planetary system.

The profit equation remains the organizing principle of the framework, while the additional disciplines determine the constraints, uncertainties, and dynamics governing its realization.

Part IV

65 Political Science, International Relations, and Geopolitics

65.1 Resource Endowments and State Formation

Throughout history, resource endowments have influenced:

- state formation,
- fiscal capacity,
- military capability,
- technological development,
- international influence.

Within the present framework, the strategic resource base of a state is

$$\Pi = \int_{r_{\max}}^R 4\pi[p(r) - c(r)]\rho(r)r^2 dr.$$

States possessing larger expected profits generally possess greater economic resources available for investment, taxation, and defense.

65.2 State Capacity

Let

$$\tau \in [0, 1]$$

denote the effective tax rate.

Government revenue generated from extraction equals

$$G = \tau\Pi.$$

Suppose state capacity is

$$C_S = f(G).$$

Then

$$\frac{\partial C_S}{\partial \Pi} = f'(G)\tau.$$

Consequently, larger extractable resource bases increase potential state capacity.

66 International Relations

66.1 Strategic Competition

Consider two states:

$$A \quad \text{and} \quad B.$$

Suppose both seek control of the same extraction region.

Expected strategic value becomes

$$V_{AB} = E[\Pi_A] + E[\Pi_B].$$

As

$$V_{AB}$$

increases, incentives for competition may increase.

66.2 Alliance Formation

Suppose states form alliances to share resource access.

Let

$$\omega_i$$

denote the allocation weight assigned to state i .

Then

$$\sum_i \omega_i = 1.$$

Alliance resource value becomes

$$\Pi_i = \omega_i \Pi.$$

Thus alliances may be interpreted as mechanisms for resource-sharing and risk-sharing.

67 Geopolitical Networks

Let

$$N = \{1, \dots, n\}$$

denote states.

Define adjacency matrix

$$A = (a_{ij}).$$

Here

$$a_{ij} = 1$$

if states interact strategically and

$$a_{ij} = 0$$

otherwise.

The resulting graph captures:

- trade relations,
- alliance structures,
- conflict structures,
- resource dependencies.

68 Warfare Economics

68.1 Military Capacity

Suppose military strength satisfies

$$M_S = \alpha + \beta \Pi.$$

Then

$$\frac{\partial M_S}{\partial \Pi} = \beta.$$

Resource extraction therefore contributes directly to military capability.

68.2 Defense Allocation

Suppose a fraction

$$\delta$$

of resource profits is allocated to defense.

Then

$$D = \delta\Pi.$$

The remaining fraction

$$1 - \delta$$

is allocated to civilian uses.

68.3 Conflict Payoffs

Suppose conflict over a resource region yields expected value

$$V.$$

Conflict cost equals

$$K.$$

Then expected net gain is

$$V - K.$$

Conflict becomes economically attractive whenever

$$V > K.$$

69 Game-Theoretic Resource Competition

Consider two players:

$$A \quad \text{and} \quad B.$$

Each chooses either:

- Cooperate (C),
- Compete (D).

The payoff matrix is

	C	D
C	(R, R)	(S, T)
D	(T, S)	(P, P)

Table 4: Payoff matrix.

Resource competition may therefore be analysed using standard game-theoretic tools.

70 Computational Architecture

70.1 Layered Planetary Model

Earth is partitioned into layers:

$$0 = r_0 < r_1 < \dots < r_n = R.$$

For layer i :

- density: ρ_i ,
- price: p_i ,
- cost: c_i .

Mass of layer i :

$$M_i = \frac{4\pi}{3} \rho_i (r_i^3 - r_{i-1}^3).$$

Profit contribution:

$$\Pi_i = (p_i - c_i) M_i.$$

Total profit:

$$\Pi = \sum_{i=1}^n \Pi_i.$$

71 Pseudo-Code

71.1 Deterministic Extraction

Input:

rho_i

p_i

c_i

r_i

Profit = 0

For i = 1, ..., n

Mass_i = (4*pi/3) rho_i (r_i^3 - r_{(i-1)}^3)

Profit_i = (p_i - c_i) Mass_i

Profit += Profit_i

End

Return Profit

71.2 Monte Carlo Extraction

Input:

N simulations

For k=1, ..., N

Draw geological parameters

Draw prices

Draw costs

Compute Profit_k

End

ExpectedProfit = Average(Profit_k)

72 Artificial Intelligence Architecture

72.1 State Space

Define state

$$s_t = (\rho_t, p_t, c_t, E_t).$$

The state contains:

- geological information,
- economic information,
- environmental information.

72.2 Actions

Actions include:

- extraction,
- exploration,
- conservation,
- technology investment.

Denote action by

$$a_t.$$

72.3 Reward Function

Reward equals

$$R_t = \Pi_t - \lambda E_t.$$

The parameter

$$\lambda$$

measures environmental penalties.

72.4 Optimal Policy

The planner solves

$$\max_{\pi} E \left[\sum_{t=0}^{\infty} \beta^t R_t \right].$$

73 Networked Planetary Resource Systems

Suppose multiple extraction regions exist.

Let

$$\Pi_i$$

denote profit from region i .

Total planetary profit becomes

$$\Pi_P = \sum_{i=1}^N \Pi_i.$$

The framework therefore scales naturally from local mines to planetary resource systems.

74 Integrated Theorem

Theorem 6 (Integrated Planetary Resource Theorem). *Suppose:*

1. *Earth possesses density field $\rho(r)$;*
2. *prices are given by $p(r)$;*
3. *costs are given by $c(r)$;*
4. *extraction occurs from $r_{\max}$ to R .*

Then total profit satisfies

$$\Pi = \int_{r_{\max}}^R 4\pi[p(r) - c(r)]\rho(r)r^2 dr.$$

Moreover,

$$E[\Pi] = \int_{r_{\max}}^R 4\pi E[(p - c)\rho]r^2 dr.$$

Proof. Combine the General Extractable Mass Theorem from Part I with the Value Decomposition Theorem from Part II and the stochastic expectation framework from Part III.

□

75 Master Table

Symbol	Meaning
R	Planetary radius
$\rho(r)$	Density field
M	Planetary mass
$r_{\max}$	Remaining core radius
M_{ext}	Extractable mass
$p(r)$	Price field
$c(r)$	Cost field
$m(r)$	Margin field
Π	Profit
$E[\Pi]$	Expected profit
G	Government revenue
D	Defense expenditure
M_S	Military strength
B	Biodiversity
$H(G)$	Geological entropy

Table 5: Unified notation across Parts I–IV.

76 PGF/TikZ Figure: Layered Earth

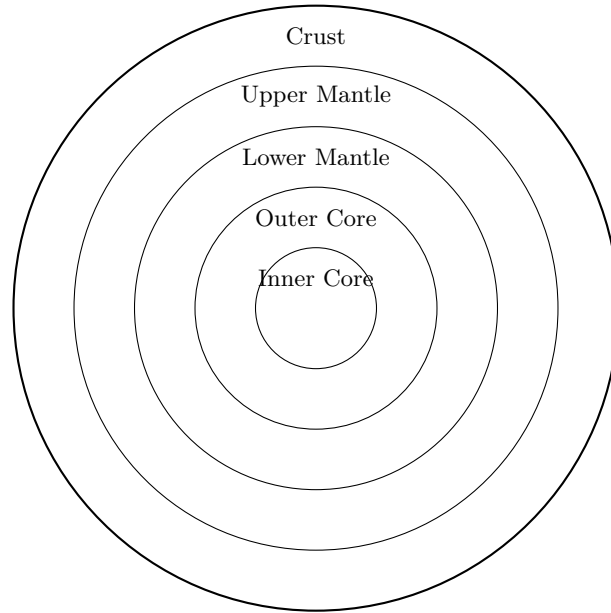


Figure 2: Layered Earth

77 Research Directions

Potential extensions include:

1. endogenous technological progress,
2. stochastic geopolitical shocks,
3. climate dynamics,
4. planetary colonization,
5. asteroid mining,
6. reinforcement-learning resource planners,
7. multi-agent resource competition,
8. world-model simulation systems.

78 General Conclusion

The complete framework developed across Parts I–IV begins with planetary geometry and culminates in an integrated stochastic resource economy.

The central equation of the framework is

$$\Pi = \int_{r_{\max}}^R 4\pi[p(r) - c(r)]\rho(r)r^2 dr.$$

This expression links:

- geometry,
- geology,
- physics,

- chemistry,
- biology,
- economics,
- finance,
- welfare economics,
- political science,
- international relations,
- warfare economics,
- statistics,
- computer science,
- artificial intelligence.

The framework may therefore be viewed as a unified mathematical architecture for planetary resource extraction under uncertainty.

Part V

79 Dynamic Optimization and Planetary Resource Control

79.1 Introduction

Parts I–IV developed a static and stochastic framework for planetary resource extraction.

The next step is to study optimal decision-making through time.

The central question becomes:

How should a planetary planner allocate extraction, exploration, conservation, technological investment, and strategic reserves in order to maximize long-run objectives?

The resulting framework naturally connects resource economics with dynamic programming, optimal control theory, stochastic processes, Bayesian learning, and artificial intelligence.

80 Dynamic Resource Stocks

80.1 Remaining Resource Stock

Let

$$S(t)$$

denote the remaining extractable stock at time t .

Extraction rate is

$$q(t) \geq 0.$$

The stock dynamics satisfy

$$\frac{dS}{dt} = -q(t).$$

Integrating,

$$S(t) = S(0) - \int_0^t q(u) du.$$

80.2 Finite Exhaustion Time

Let

$$T_E$$

denote exhaustion time.

Then

$$S(T_E) = 0.$$

Hence

$$S(0) = \int_0^{T_E} q(t) dt.$$

81 Capital Accumulation

Let

$$K(t)$$

denote extraction capital.

Examples include:

- mines,
- drilling equipment,
- transportation systems,
- refineries,
- AI infrastructure.

Capital evolves according to

$$\frac{dK}{dt} = I(t) - \delta K(t),$$

where

- $I(t)$ is investment,
- δ is depreciation.

82 Technology Dynamics

Let

$$A(t)$$

denote technological capability.

Assume

$$\frac{dA}{dt} = \eta R_A(t),$$

where

$$R_A(t)$$

denotes research expenditure.

Improved technology may reduce extraction costs.

Thus

$$\frac{\partial c}{\partial A} < 0.$$

83 Dynamic Profit Function

Instantaneous profit becomes

$$\pi(t) = \int_{r_{\max}(t)}^R 4\pi[p(r, t) - c(r, t)]\rho(r)r^2 dr.$$

Total discounted profit equals

$$J = \int_0^\infty e^{-\rho_D t} \pi(t) dt,$$

where

$$\rho_D > 0$$

is the discount rate.

84 Optimal Control Problem

Definition 5. *The planetary planner solves*

$$\max_{q(t), I(t)} J.$$

Subject to

$$\frac{dS}{dt} = -q(t),$$

and

$$\frac{dK}{dt} = I(t) - \delta K(t).$$

85 Hamiltonian Formulation

Define costate variables

$$\lambda_S(t),$$

and

$$\lambda_K(t).$$

The Hamiltonian becomes

$$H = \pi(t) + \lambda_S(-q) + \lambda_K(I - \delta K).$$

Theorem 7 (Necessary Conditions). *An interior optimum satisfies*

$$\frac{\partial H}{\partial q} = 0,$$

$$\frac{\partial H}{\partial I} = 0,$$

together with the costate equations

$$\dot{\lambda}_S = \rho_D \lambda_S - \frac{\partial H}{\partial S},$$

$$\dot{\lambda}_K = \rho_D \lambda_K - \frac{\partial H}{\partial K}.$$

86 Bellman Equation

86.1 Value Function

Let

$$V(S, K, A, t)$$

denote the value function.

The Bellman equation is

$$\rho_D V = \max_{q, I} \{ \pi + V_S(-q) + V_K(I - \delta K) + V_A \eta R_A \}.$$

Remark 2. *The Bellman equation provides a recursive representation of the optimization problem.*

87 Stochastic Prices

Suppose commodity prices evolve according to

$$dp_t = \mu_p p_t dt + \sigma_p p_t dW_t,$$

where

$$W_t$$

is standard Brownian motion.

Definition 6. *The process*

$$dp_t = \mu_p p_t dt + \sigma_p p_t dW_t$$

is geometric Brownian motion.

88 Stochastic Costs

Extraction costs may also fluctuate.

Assume

$$dc_t = \mu_c c_t dt + \sigma_c c_t dZ_t,$$

where

$$Z_t$$

is Brownian motion.

Then margins become stochastic.

89 Stochastic Resource Dynamics

Suppose geological discoveries occur.

Then

$$dS_t = [-q_t + \mu_S]dt + \sigma_S dW_t.$$

Here

- μ_S captures discoveries,
- σ_S captures uncertainty.

90 Hamilton–Jacobi–Bellman Equation

The stochastic Bellman equation becomes

$$\rho_D V = \max_a \{ \pi + \mathcal{L}V \},$$

where

$$\mathcal{L}$$

is the infinitesimal generator.

Explicitly,

$$\mathcal{L}V = V_S \mu_S + \frac{1}{2} \sigma_S^2 V_{SS} + \dots.$$

This equation governs optimal planetary extraction under uncertainty.

91 Bayesian Learning

91.1 Unknown Geological Parameters

Suppose parameter

$$\theta$$

is unknown.

Prior beliefs satisfy

$$p(\theta).$$

After observing data

$$D,$$

beliefs update according to

$$p(\theta|D) = \frac{p(D|\theta)p(\theta)}{p(D)}.$$

Theorem 8 (Bayes' Rule). *Posterior beliefs equal prior beliefs multiplied by likelihood and normalized.*

92 Exploration as Information Acquisition

Let

$$I_G$$

denote geological information.

Suppose exploration reduces uncertainty.

Then

$$H(G|I_G) < H(G),$$

where

$$H$$

is entropy.

Exploration therefore possesses option value.

93 Real Options

Suppose extraction may be delayed.

Define option value

$$\Omega.$$

Then project value becomes

$$F = NPV + \Omega.$$

Uncertainty may therefore increase project value.

94 Markov Decision Processes

Define state

$$s_t.$$

Action

$$a_t.$$

Transition probability

$$P(s_{t+1}|s_t, a_t).$$

Reward

$$R(s_t, a_t).$$

The planetary extraction problem becomes a Markov Decision Process.

95 Reinforcement Learning

95.1 Q-Function

Define

$$Q(s, a).$$

Bellman optimality becomes

$$Q^*(s, a) = E \left[R + \beta \max_{a'} Q^*(s', a') \right].$$

96 Deep Reinforcement Learning

Neural networks approximate

$$Q_\theta(s, a).$$

Training minimizes

$$L(\theta) = E \left[(y - Q_\theta)^2 \right].$$

where

$$y = R + \beta \max_{a'} Q_{\theta^-}(s', a').$$

97 Multi-Agent Reinforcement Learning

Suppose multiple states compete.

Agents:

$$A_1, \dots, A_n.$$

Each solves

$$\max E \left[\sum_t \beta^t R_t^i \right].$$

Strategic interactions generate:

- cooperation,
- competition,

- alliances,
- conflict.

98 World Models

Let latent state be

$$z_t.$$

World dynamics satisfy

$$z_{t+1} = F(z_t, a_t, \varepsilon_t).$$

The planner learns:

- geology,
- economics,
- environmental effects,
- geopolitical reactions.

through simulation.

99 Digital Twin of Earth

A planetary digital twin consists of:

1. geological model,
2. economic model,
3. environmental model,
4. geopolitical model,
5. AI planning model.

The digital twin generates forecasts of

$$\Pi,$$

$$W,$$

and

$$S.$$

under alternative policies.

100 Planetary Resource Operating System

The framework may be interpreted as a resource operating system.

Inputs:

- geological data,
- market data,
- environmental data,
- strategic data.

Outputs:

- extraction plans,
- investment plans,
- reserve policies,
- geopolitical recommendations.

101 Unified Dynamic Resource Equation

Combining Parts I–V yields

$$\Pi_t = \int_{r_{\max}(t)}^R 4\pi[p(r, t) - c(r, t)]\rho(r, t)r^2 dr.$$

State evolution satisfies

$$dS_t = [-q_t + \mu_S]dt + \sigma_S dW_t.$$

Optimal policy satisfies

$$\rho_D V = \max_a \{\pi + \mathcal{L}V\}.$$

Together these equations characterize extraction, uncertainty, learning, and control.

102 Algorithmic Framework

102.1 High-Level Pseudo-Code

Initialize beliefs

Observe geology

Update posterior

Estimate resource value

Solve Bellman equation

Choose extraction policy

Observe outcomes

Update beliefs

Repeat

103 Conclusion

Part V transformed the heterogeneous-Earth extraction model into a dynamic decision system.

The resulting framework unifies:

- optimal control,
- dynamic programming,
- stochastic differential equations,

- Bayesian inference,
- real options,
- reinforcement learning,
- multi-agent systems,
- world models,
- planetary digital twins.

The Earth is therefore represented not merely as a geological object but as a stochastic dynamical system whose resources may be managed through recursive optimization, probabilistic learning, and intelligent control.

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Glossary

- R Planetary radius.
- M Planetary mass.
- $\rho(r)$ Density field.
- $r_{\max}$ Radius of deepest remaining core.
- M_{ext} Extractable mass.
- M_{core} Remaining mass.
- κ Accessibility coefficient.
- $v(r)$ Value density.
- V_{ext} Recoverable economic value.
- Ω Probability space.
- $\rho(r)$ Density field.
- $p(r)$ Price field.

- $c(r)$ Cost field.
- $m(r)$ Margin field.
- Π Profit.
- $E[\Pi]$ Expected profit.
- G Government revenue.
- D Defense expenditure.
- M_S Military strength.
- $\chi(r)$ Resource concentration field.
- $\gamma(r)$ Recovery efficiency.
- $H(G)$ Geological entropy.
- s_t State variable in AI framework.
- a_t Action variable.
- R_t Reward function.
- β Discount factor.

The End